

1 Solution — GIAM 5.3

Problem 8

1.1 Problem

Give an inductive proof that $\forall n \in \mathbb{N}, 7 \mid 8^n + 6$.

1.2 Answer

$$8^{k+1} + 6 = 8(8^k + 6) - 42$$

Both terms on the right are multiples of 7 once $8^k + 6$ is, since $42 = 7 \cdot 6$. Verified for every n up to 3000.

1.3 The Proof

Theorem. For every natural number n , 7 divides $8^n + 6$.

Proof. By induction on n .

Basis. At $n = 0$, $8^0 + 6 = 7$, and $7 \mid 7$.

Inductive step. Let k be a particular but arbitrarily chosen natural number and suppose $8^k + 6 = 7m$ for some integer m . Then

$$\begin{aligned} 8^{k+1} + 6 &= 8 \cdot 8^k + 48 - 42 \\ &= 8(8^k + 6) - 42 \\ &= 8 \cdot 7m - 7 \cdot 6 \\ &= 7(8m - 6). \end{aligned}$$

Since $8m - 6$ is an integer, $7 \mid 8^{k+1} + 6$.

Basis and step together give the result for every n . ■

The move is the reverse of 5.3 Problem 7's. There we split $13 \cdot 13^k$ to find $f(k)$ inside $f(k+1)$; here we multiply the whole hypothesis by 8 and correct the constant, adding $48 - 42$ in place of 6. The additive route works equally well: $(8^{k+1} + 6) - (8^k + 6) = 7 \cdot 8^k$.

1.4 The Picture

Base 8 makes the divisibility visible at a glance.

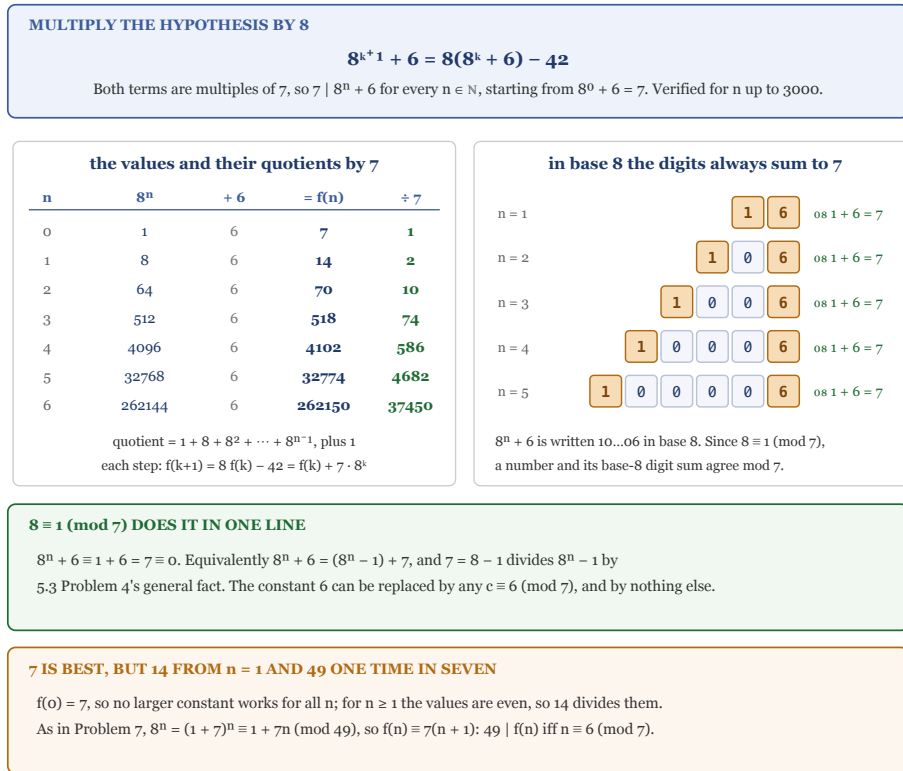


Figure 1.1: A diagram in four parts. The top band states the step identity eight to the k plus one, plus six, equals eight times the quantity eight to the k plus six, minus forty-two. The left panel tabulates, for n from zero to six, eight to the n plus six: 7, 14, 70, 518, 4102, 32774, 262150, and the quotients by seven: 1, 2, 10, 74, 586, 4682, 37450, each equal to one plus the geometric series one plus eight up to eight to the n minus one. The right panel writes eight to the n plus six in base eight for n from one to five, as 16, 106, 1006, 10006, 100006, whose digits always sum to seven. A green band gives the one-line proof from eight congruent to one mod seven. An orange band notes that

seven is the best constant, fourteen divides from n equal to one on, and forty-nine divides exactly when n is congruent to six mod seven.

1.5 Remark — The One-Line Congruence Proof

Since $8 \equiv 1 \pmod{7}$, every power of 8 is $\equiv 1$, and

$$8^n + 6 \equiv 1 + 6 = 7 \equiv 0 \pmod{7}.$$

Put another way, $8^n + 6 = (8^n - 1) + 7$, and $7 = 8 - 1$ divides $8^n - 1$ by the general fact $a - 1 \mid a^n - 1$ from 5.3 Problem 4. The quotient is

$$\frac{8^n + 6}{7} = (1 + 8 + 8^2 + \cdots + 8^{n-1}) + 1,$$

verified for n up to 500. The constant 6 was the only choice that works, up to multiples of 7: a search over c from 0 to 13 finds that $7 \mid 8^n + c$ for every n only for $c = 6$ and $c = 13$.

1.6 Remark — Casting Out Sevens in Octal

In base 10, $10 \equiv 1 \pmod{9}$ gives the familiar test: a number and its digit sum leave the same remainder modulo 9. In base 8, $8 \equiv 1 \pmod{7}$ gives the same test for 7; it was checked for every number below 20000. Since

$$8^n + 6 = 10 \cdots 06_8 \quad (n \geq 1),$$

a 1 followed by $n - 1$ zeros and a 6, its digit sum is $1 + 6 = 7$, and divisibility by 7 is immediate. The problem is the octal version of noticing that $10^n + 8$ is always divisible by 9.

The quotients have the matching form $11 \cdots 12_8$: for instance $4682 = 11112_8$, which is $11111_8 = 1 + 8 + 8^2 + 8^3 + 8^4$ plus 1.

1.7 Remark — Is 7 the Best Divisor?

As a constant valid for every n , yes, since $8^0 + 6 = 7$. But the basis is the only obstacle to more: for $n \geq 1$ every value is even, so $14 \mid 8^n + 6$, and the gcd of the values for

$n = 1, \dots, 299$ is 14.

Higher powers of 7 follow the pattern of 5.3 Problem 7. Writing $8 = 1 + 7$, the binomial theorem gives $8^n \equiv 1 + 7n \pmod{49}$, so

$$8^n + 6 \equiv 7 + 7n = 7(n + 1) \pmod{49}.$$

Hence $49 \mid 8^n + 6$ exactly when $n \equiv 6 \pmod{7}$; then 98 divides as well, since such n are at least 6. Both were checked to $n = 3000$. The first instance is $8^6 + 6 = 262150 = 98 \cdot 2675$.